



Tag-along

GROUP-13 SOFTWARE

Mentors:

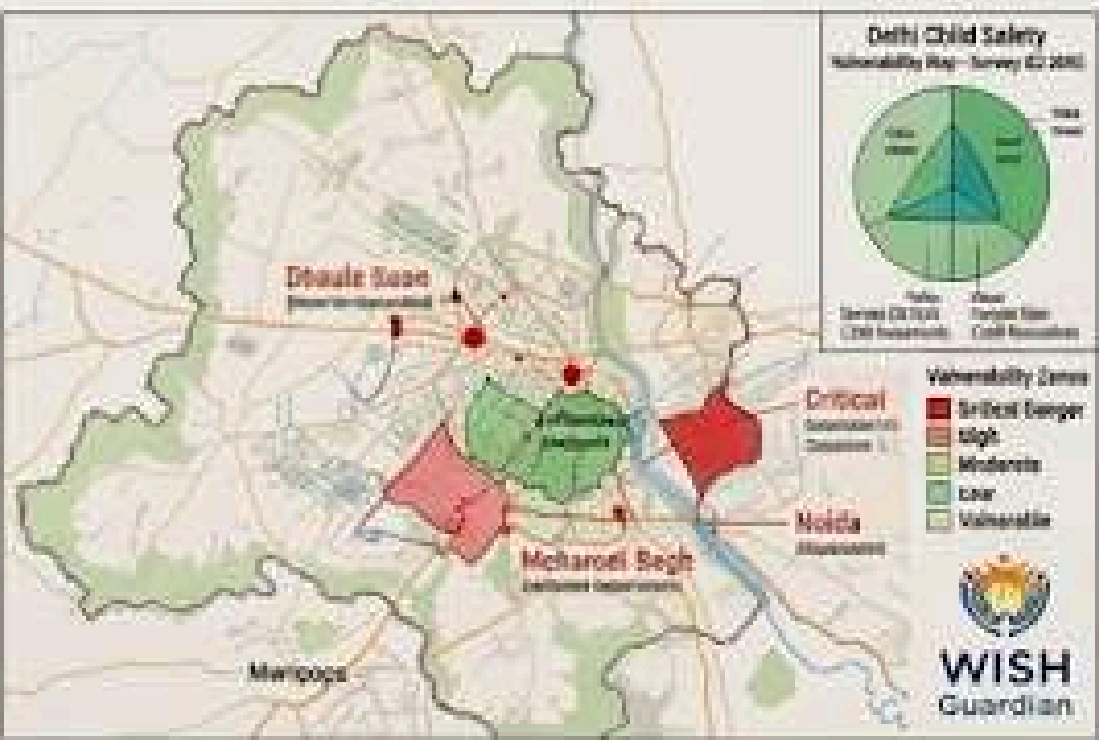
**Neha Francis
Mudit Bhansali**



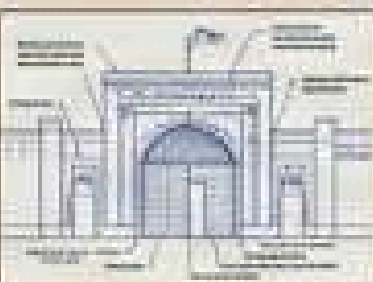
Presenters:

**Diya Dibyanshi
Ishita Sahoo
Pankhudi Pandey
Tanushree Gupta
Vidushi Kochhar**

CHILD SAFETY AWARENESS NEEDS



The 1978 Case That Shook Delhi: Geeta & Sanjay Chopra.



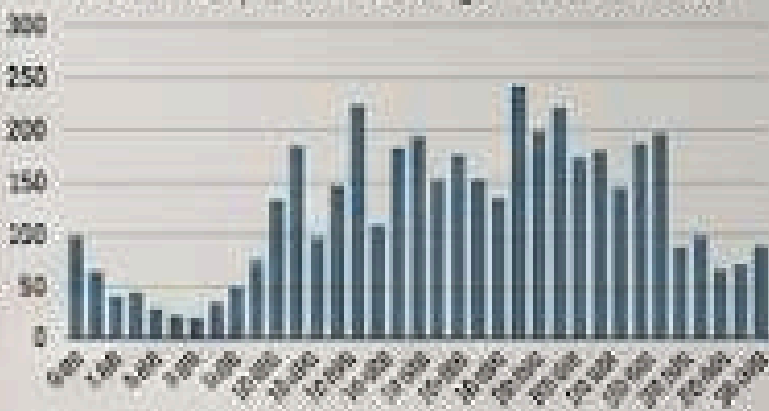
Tihar Jail - Historical Entrance Schematic, circa 1982.

The Geeta and Sanjay Chopra case (1978, Delhi) is probably the most infamous. The siblings accepted a ride from strangers near Dhaura Kuan, and their abductors, Kuljeet Singh (alias Ranga) and Jasbir Singh (alias Billa) (Wikipedia), killed them, believing the family was wealthy.

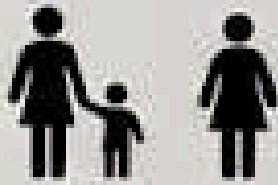
The kidnappers were arrested on a train in September 1978 and, after trial, were hanged at Tihar Jail in January 1982 (Stillunfol). The case led to lasting changes in how Delhi treats child safety awareness, and bravery awards for children were named after the victims.

**KIDNAPPING PATTERNS.

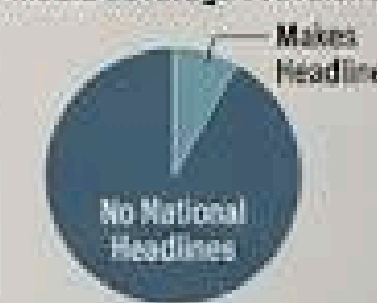
Abductions per Hour (Average India, 2015 Data)



Key Stat: ~10 People Abducted Every Hour in India, 6 are Children



Media Coverage Prevalence



Sources: Penguin Random House Research, WISH Guardian Internal Survey. *Context is crucial for the WISH Guardian project.

Hindustan Times

Ransom Kidnapping: The 2006 Anant Gupta Case, Noida.

The Anant Gupta case (2005, Noida) is a frequently cited ransom kidnapping. Three-year-old Anant, son of the Adobe India CEO, was taken from outside his home by two men on a motorcycle while his maid was walking him to the school bus (Stillunfol). A ransom call demanding Rs. 50 lakh came sa that evening, and police conducted a joint operation, eventually recovering the ransom money and rescuing the child (Stillunfol).

Political Abduction: The 1989 Kidnapping of Rubaiya Sayeed.

The Rubaiya Sayeed case (1989, 18K) was politically significant rather than ransom-related. She was the daughter of Mufti Mohammad Sayeed, then Union Home Minister, and was abducted by JKLF militants in December 1989 (Yahool), her release was negotiated in exchange for freeing five militants, a controversial decision often cited as emboldening militancy in Kashmir.

THE TIMES OF INDIA

NEW

Apr 05, 2016

Domestic Help Risk: The 2011 Ishaan Singh Case, South Delhi.

The Ishaan Singh case (2011, Delhi) is notable for highlighting domestic-help-related risk. The 18-month-old was taken by his family's domestic servant from a park in Maharani Bagh, South Delhi (Stillunfol), which made many households more cautious about staff vetting.

दैनिक भास्कर

GEETA & SANJAY CHOPRA: LASTING CHANGES IN DELHI. BRAVERY AWARDS FOR CHILDREN.

PROBLEM STATEMENT

Children in rural India are under high risk of being kidnapped, getting lost, and distress during daily commutes.

Long battery life

Network Coverage

Pocket friendly

CORE CHALLENGES



DESIGN THINKING MAP

W+SH



Step 1

Empathize



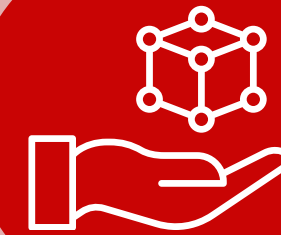
Step 2

Define



Step 3

Ideate



Step 4

Prototype



Step 5

Test

EMPATHIZE

What if the wristband is spotted and removed?

Will my child remember to press SOS in a scary situation?

What if I forget to charge the device?

Looks like a lot of problems, alva?

We've got a solution for every one of them.





Step 2

Define

Who are we competing with?

Bluetooth Trackers

Apple AirTag 2

₹2,450 • No monthly fee

Uses 2 BILLION iPhones worldwide as a hidden radar — if any iPhone is near your lost item, you get its location.

Samsung SmartTag 2

₹1,680–₹2,520 • No monthly fee

Lasts 700 days on one battery — longest in the market. Tight Samsung phone integration.

Tile Pro

₹2,940 • Free / ₹252/mo (Premium)

Works on both iPhone and Android — the only cross-platform option. Has a hidden SOS panic alert.

Chipolo Pop

₹2,450 • No monthly fee

Works on BOTH Apple and Google networks simultaneously — unique dual-network coverage.

GPS Trackers



₹7,559 device • ₹811/month

Plugs into your car's OBD port (under the steering wheel) — zero setup, instant tracking + crash detection.



₹1,260–₹2,520 • ₹836–₹1,676/month

Magnetic. Stick it hidden under any metal surface. 6-month battery. Works in 155+ countries.



₹907–₹1,259 • ₹1,092–₹1,680/month

Cheapest hardware of any GPS tracker. Also uses WiFi + Bluetooth as backup when GPS signal is weak.



₹2,940–₹4,200 • ₹1,260–₹1,928/month

Updates location every 5 seconds — the fastest real-time update interval available.

Fleet Solutions



Monthly: ₹2,268–₹2,772 per vehicle/month

AI cameras watch the driver and warn them in real time — distraction, fatigue, phone use. Also gives full ELD compliance for trucking laws.



Monthly: ₹1,680–₹2,520 per vehicle/month

Runs on Verizon's own telecom network — best coverage guarantees. Auto-plans routes for fleets across entire cities.

SURVEY ANALYSIS

68,965

Avg cases / year

81%

Female victims

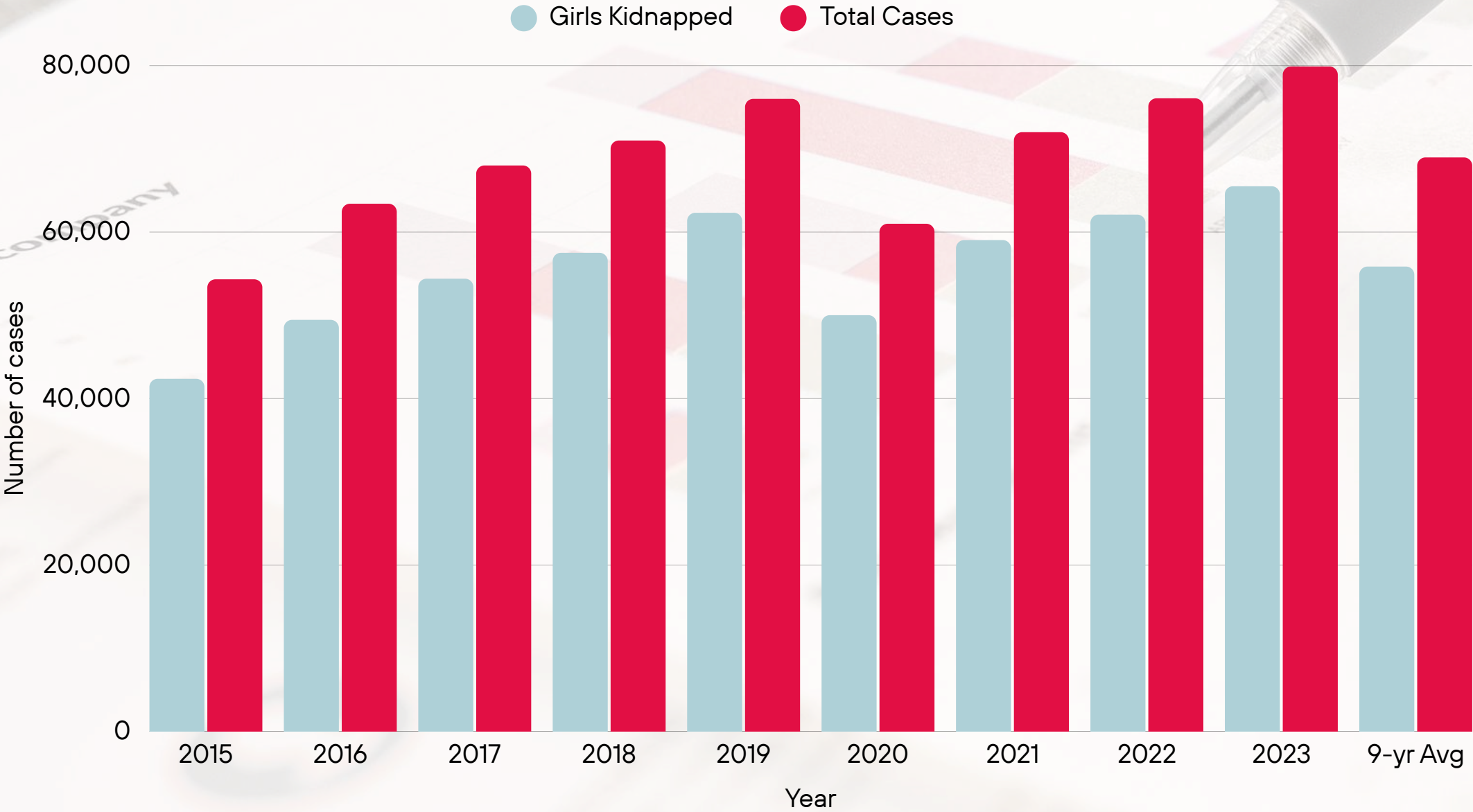
47K+

Still missing

+47%

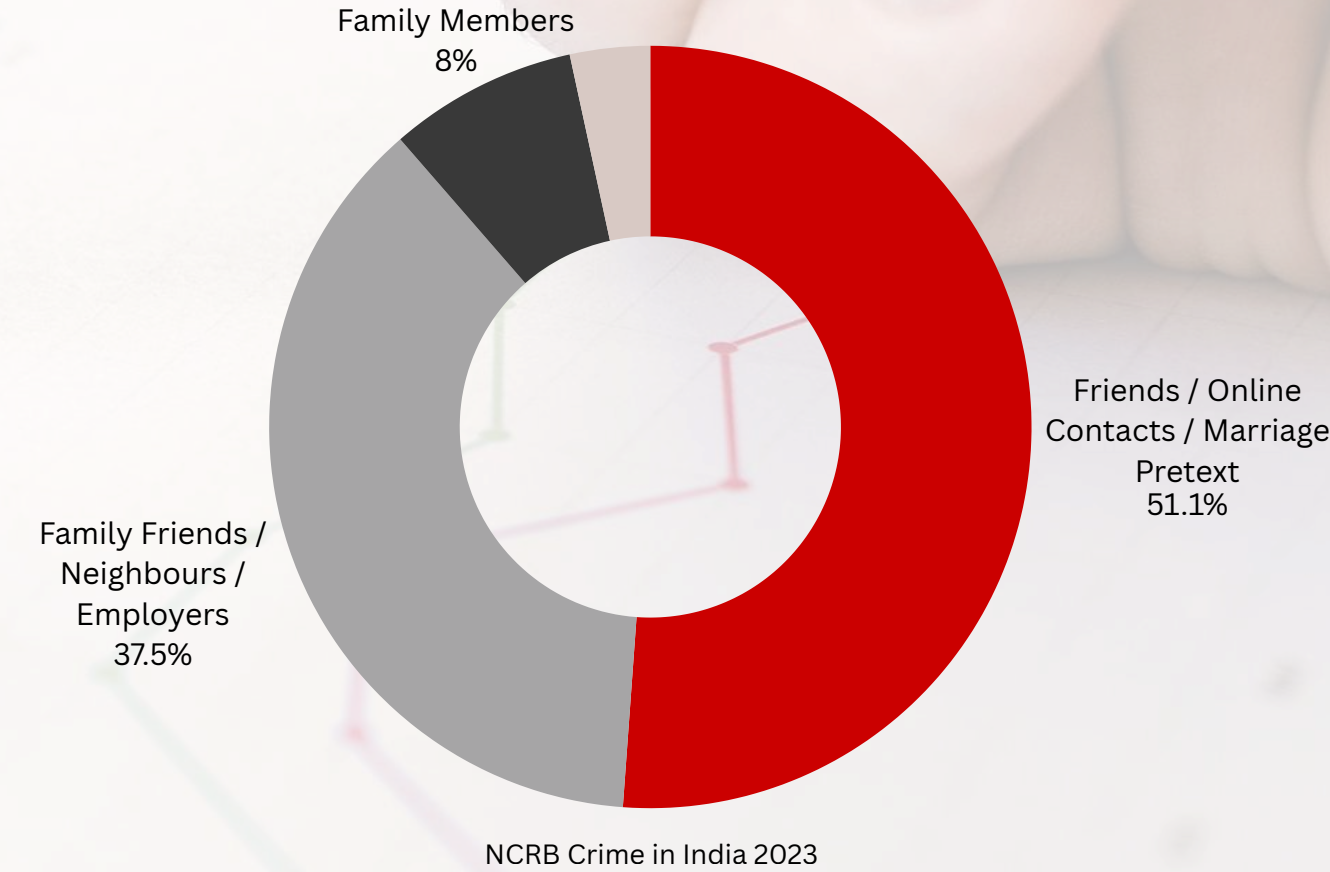
Rise 2015–2023

Child Kidnapping Cases in India (2015–2023)



National Crime Records Bureau (NCRB)

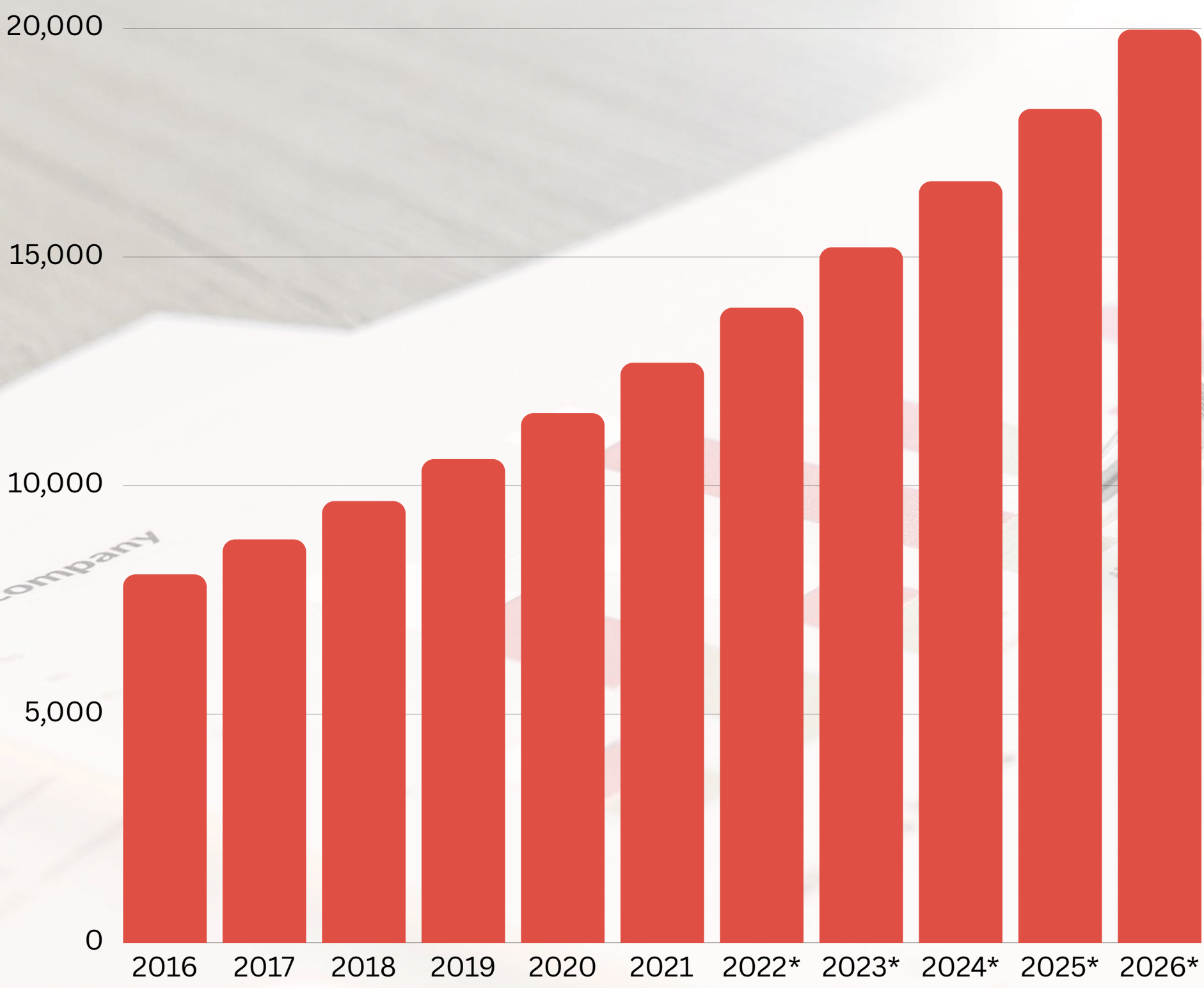
96.6% of offenders were known to the child, while only 3.4% were strangers or unidentified individuals.



NCRB Crime in India 2023

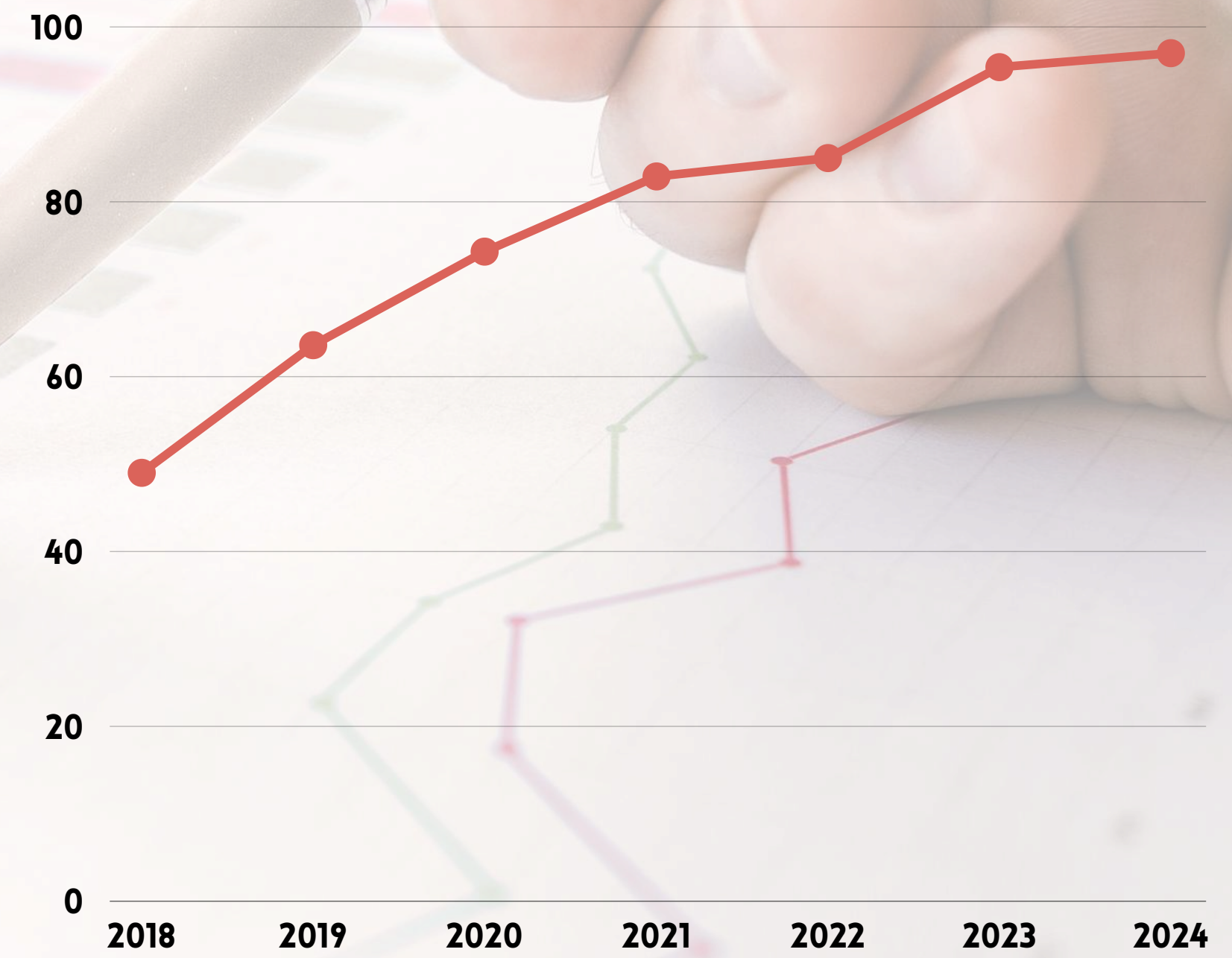
0-6 yrs	Girls Count: 620	Boys Count: 142	6-12 yrs	Girls Count: 2575	Boys Count: 654
12-16 yrs	Girls Count: 12800	Boys Count: 2644	16-18 yrs	Girls Count: 18400	Boys Count: 3011

"Rural Household Income Growth in India (2016–2026)"



NABARD NAFIS 2016-17 & 2021-22; 2022-2026 projected using NABARD-reported 9.5% CAGR.

GROWING DIGITAL CONNECTIVITY ENABLES NATIONWIDE 4G-BASED CHILD SAFETY SOLUTIONS



TRAI TELECOM STATISTICS, MINISTRY OF COMMUNICATIONS (4G VILLAGE COVERAGE), 2025-2026 VALUES PROJECTED FROM RECENT SUBSCRIBER GRO

Rural incomes increased by nearly 58% in five years while 4G connectivity reached over 95% of villages



Step 3

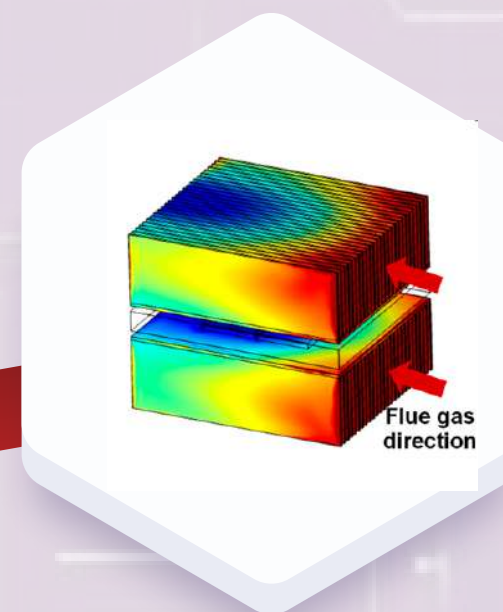
Ideate

POWER LAYER



LI ION BATTERY

- FINITE CAPACITY ✗
- INCONVENIENCE OF CHARGING ✗



TEG ENERGY HARVEST

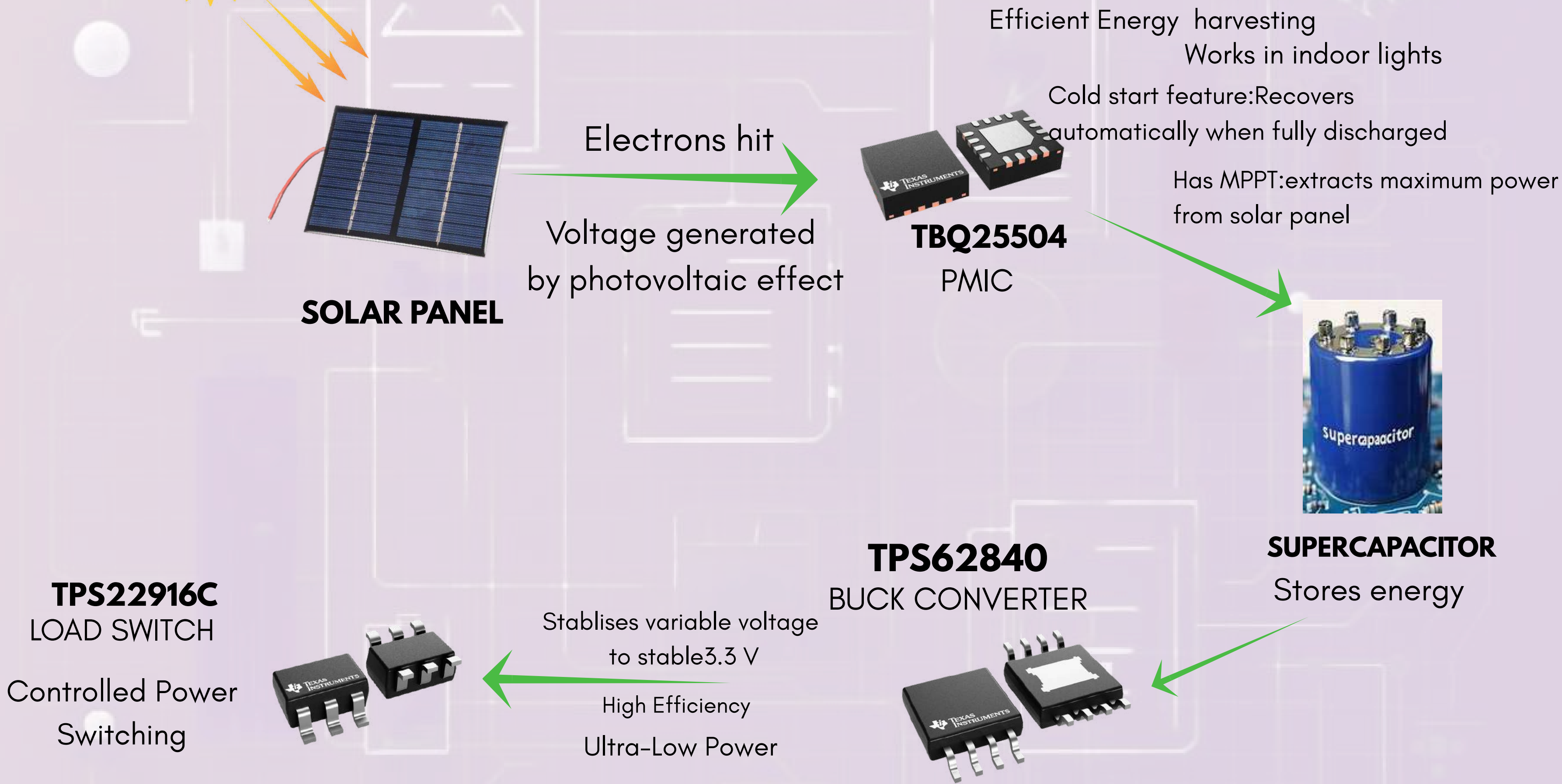
- THERMOELECTRIC GENERATION ✗
- PASSIVE ENERGY HARVEST
- FROM BODY HEAT
- PERFORMANCE DIES IN HUMID CLIMATE ✗



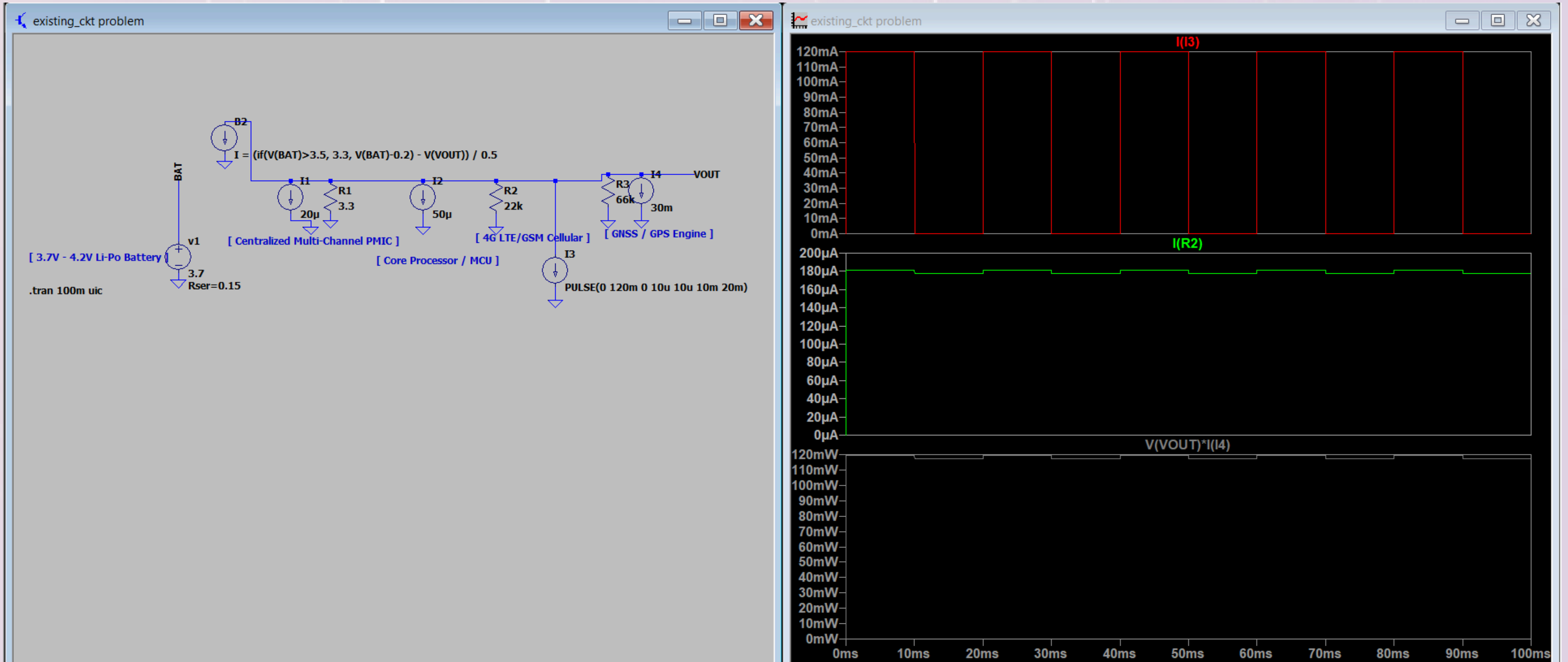
SOLAR PANEL

- RECURRING ENERGY ✓
- (LONG BATTERY LIFE)
- LOW COST ✓
- LARGE SCALE DEPLOYMENT

WORKING



LOOPHOLES IN EXISTING APPROACHES



PREVENTION OF LEAKAGE CURRENT

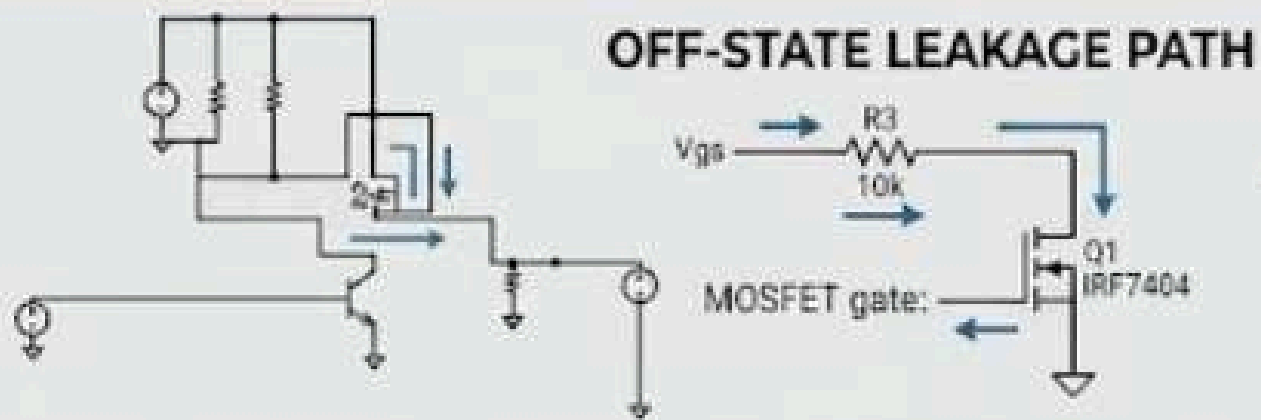
MOSFET GATING



TPS22916C MODULE



CIRCUIT A: DISCRETE MOSFET



Legacy
Discrete



OFF-STATE CURRENT
 $\approx 25 \mu\text{A}$ (Max Typ.)

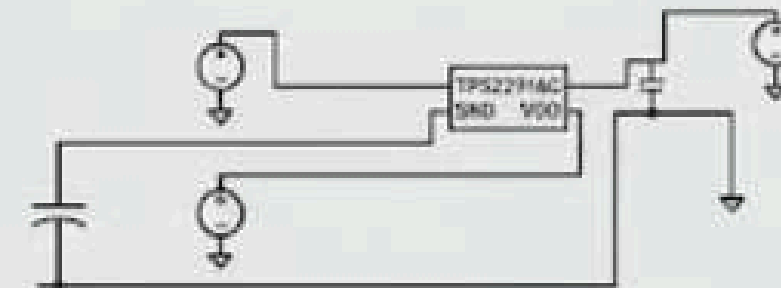


STATIC POWER LEAKAGE
 $\approx 105.00 \mu\text{W}$ (Est.)

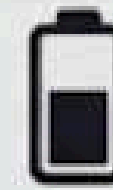
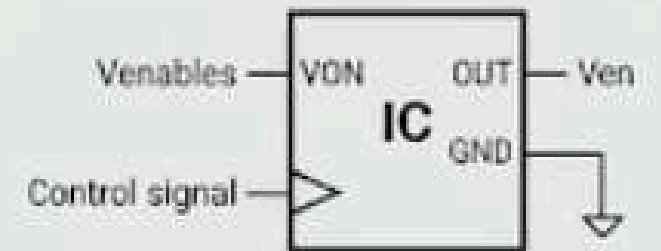
**2,500x
LEAKAGE
REDUCTION!**



CIRCUIT B: INTEGRATED LOAD SWITCH



ADVANCED INTEGRATED



OFF-STATE SHUTDOWN CURRENT
 $\approx 10 \text{ nA}$ (0.01 μA Typ.)



STATIC POWER LEAKAGE
 $\approx 0.042 \mu\text{W}$ (42 nW Typ.)

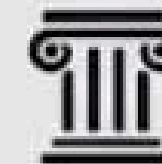


WINNER: INTEGRATED TPS22916C LOAD SWITCH



**Reduced
Standby Leakage**

**Compompact
Count**

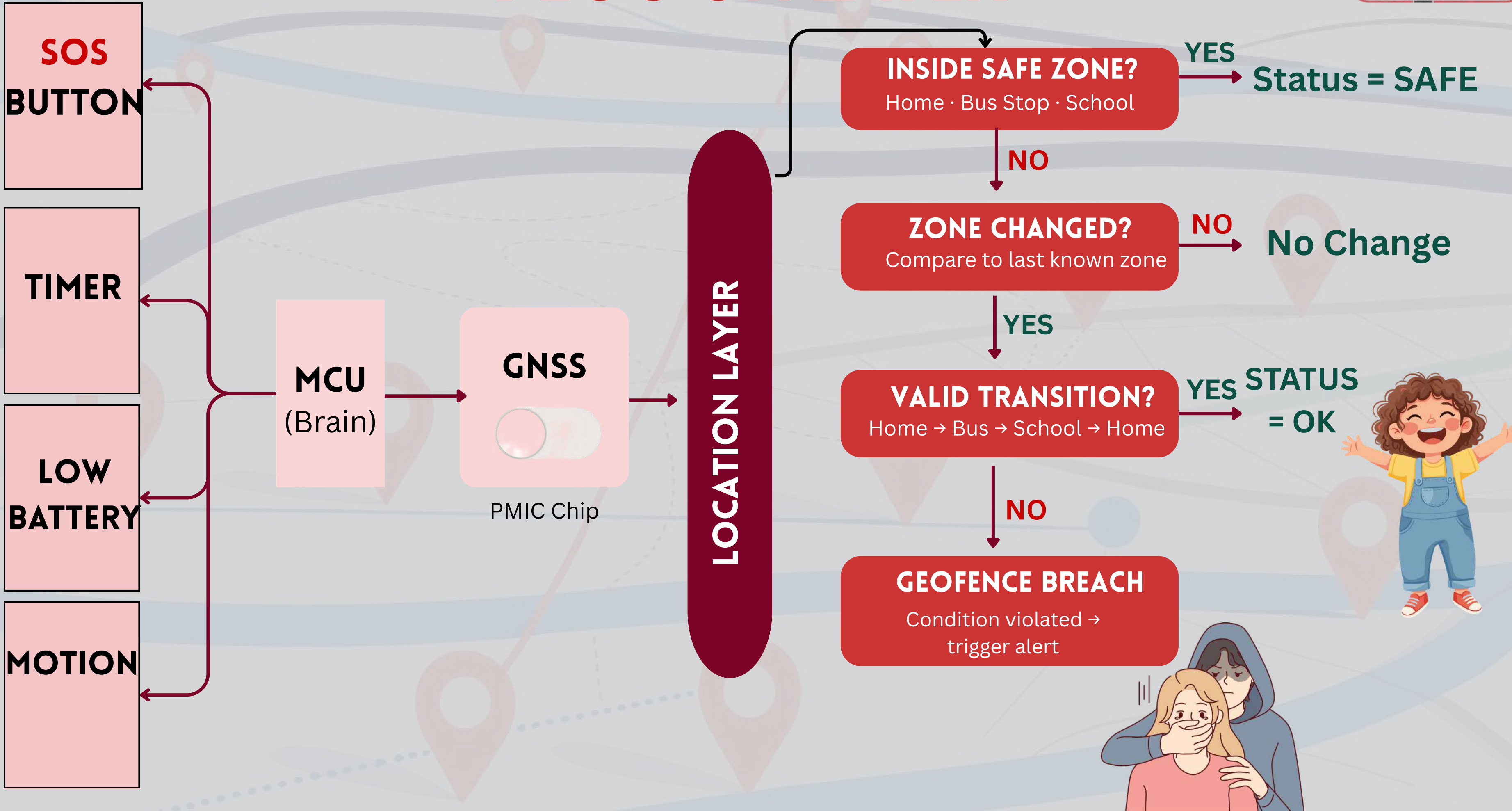


**Lower Bill
of Materials**



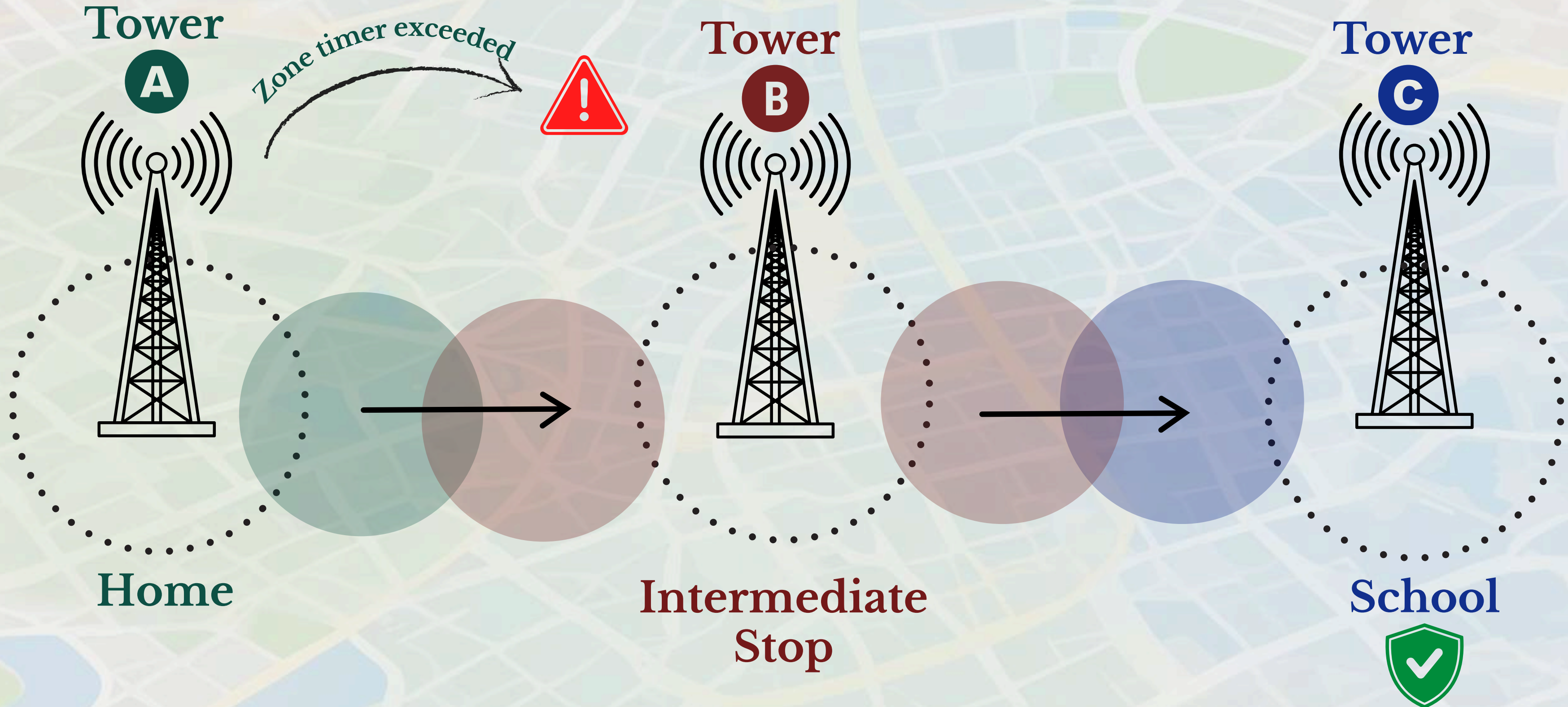
**Simplified
Design**

DECISION LAYER



ZONE MAPPING

W+SH



A one-time setup maps local Cell IDs to safe zones and links the device SIM with parent contacts, enabling autonomous monitoring and alert delivery.

NETWORK LAYER



2G (SIM800L)

- LESS COSTLY
- SLOW DATA TRANSMISSION
- WIDESPREAD SHUTDOWN



LORA-MESH NETWORK

- LONG RANGE COVERAGE
- OPERATES WITHOUT CELLULAR NETWORK
- REQUIRES LORA GATEWAYS WHICH ARE SCARCE



4G LTE CAT-1 + GNSS

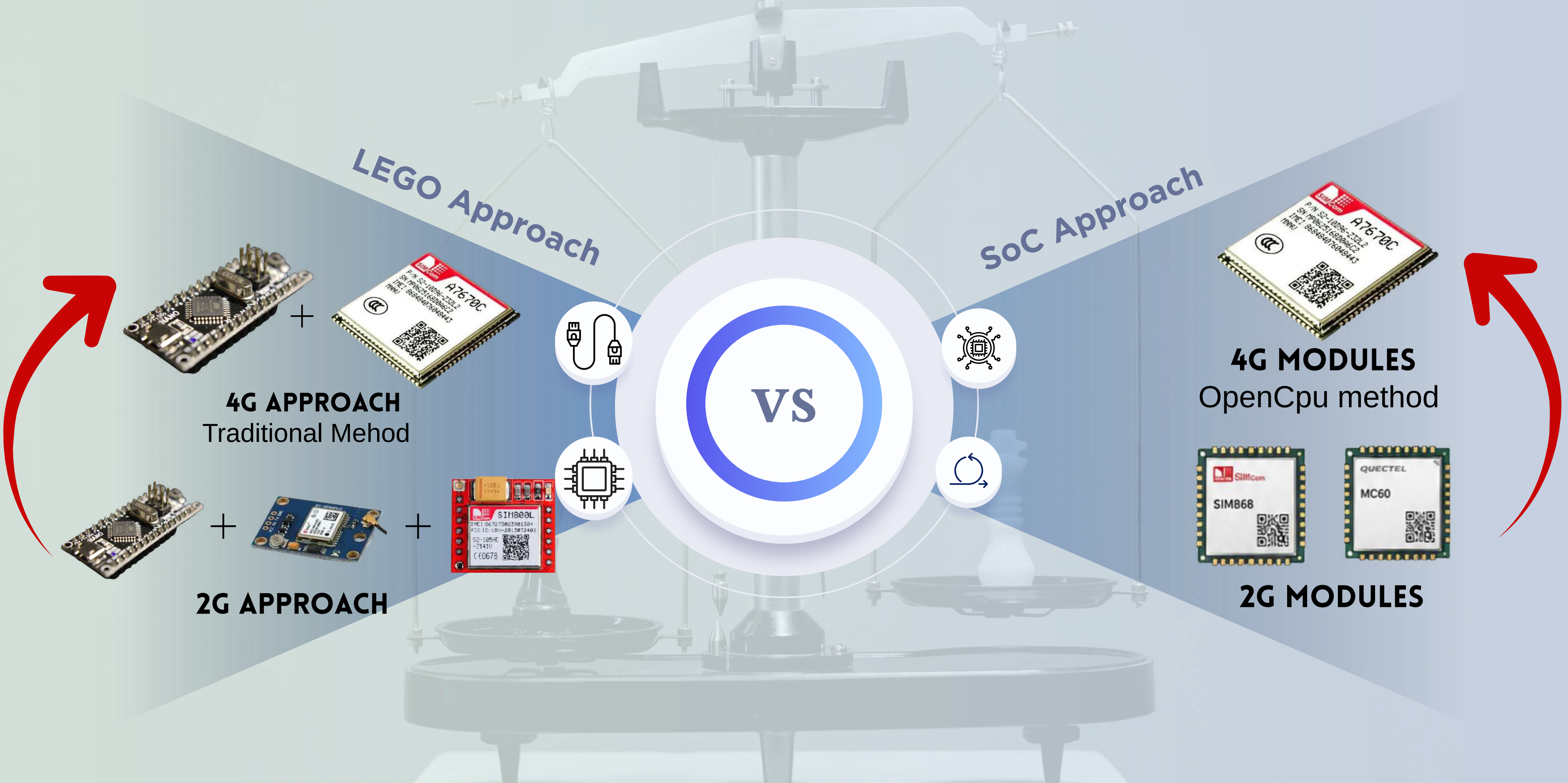
- WIDELY AVAILABLE AND USED
- GNSS SUPPORT FOR LOCATION TRACKING
- REDUCE OVERALL COMPLEXITY



Step 4

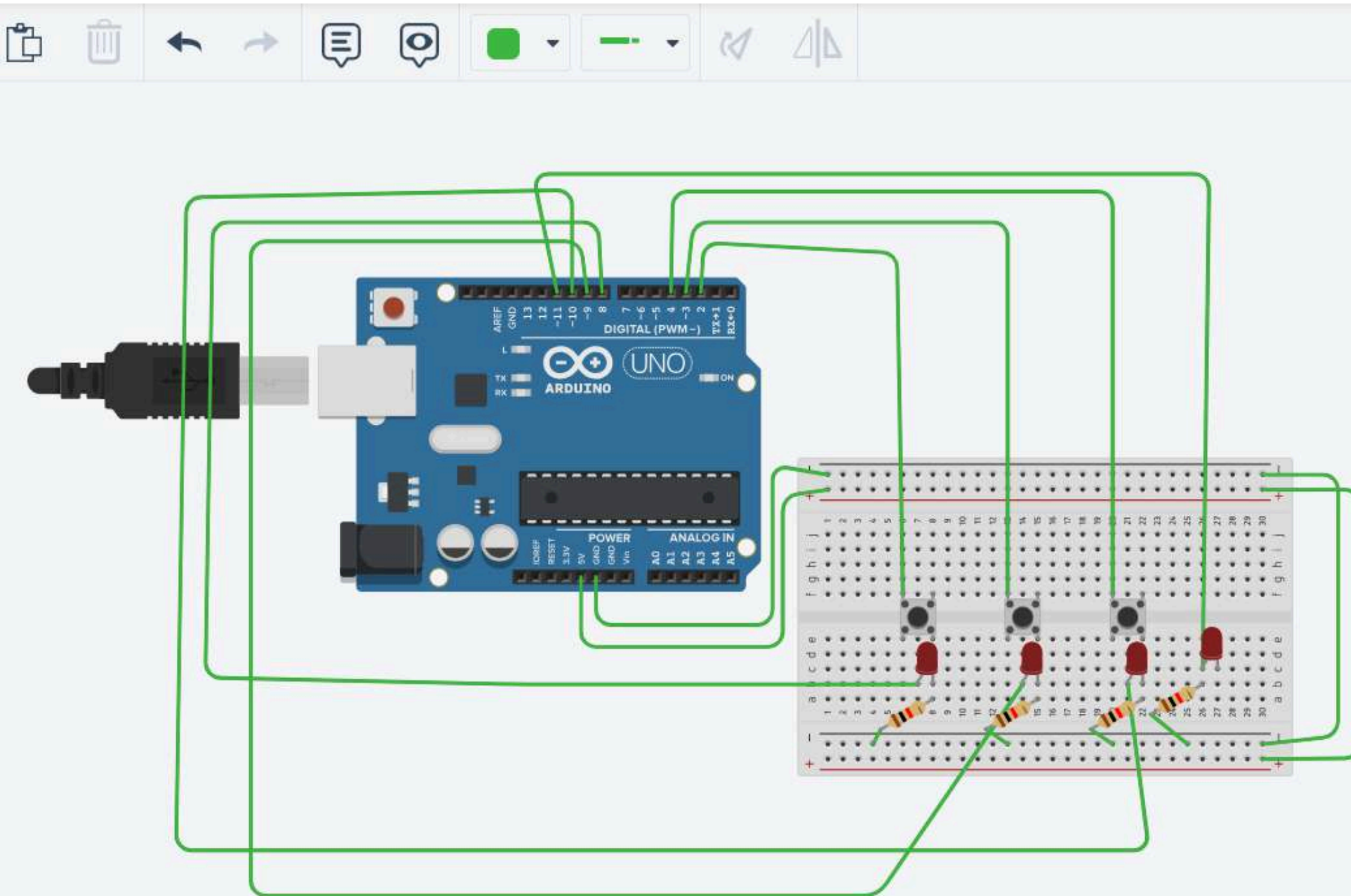
Prototype

LEGO V/S SoC



SIMULATION OF CELL TOWER

Cell ID Zone detection



Code

Start Simulation

Send

Text

1 (Arduino Uno R3)

```
103  
104     digitalWrite(homeLED, LOW);  
105     digitalWrite(busLED, LOW);
```

Serial Monitor

Cell_ID_A Detected
Current Zone : HOME
VALID TRANSITION
System Started

Cell_ID_A Detected
Current Zone : HOME
VALID TRANSITION

Cell_ID_B Detected
Current Zone : BUS STOP
VALID TRANSITION

Cell_ID_C Detected
Current Zone : SCHOOL
VALID TRANSITION
System Started

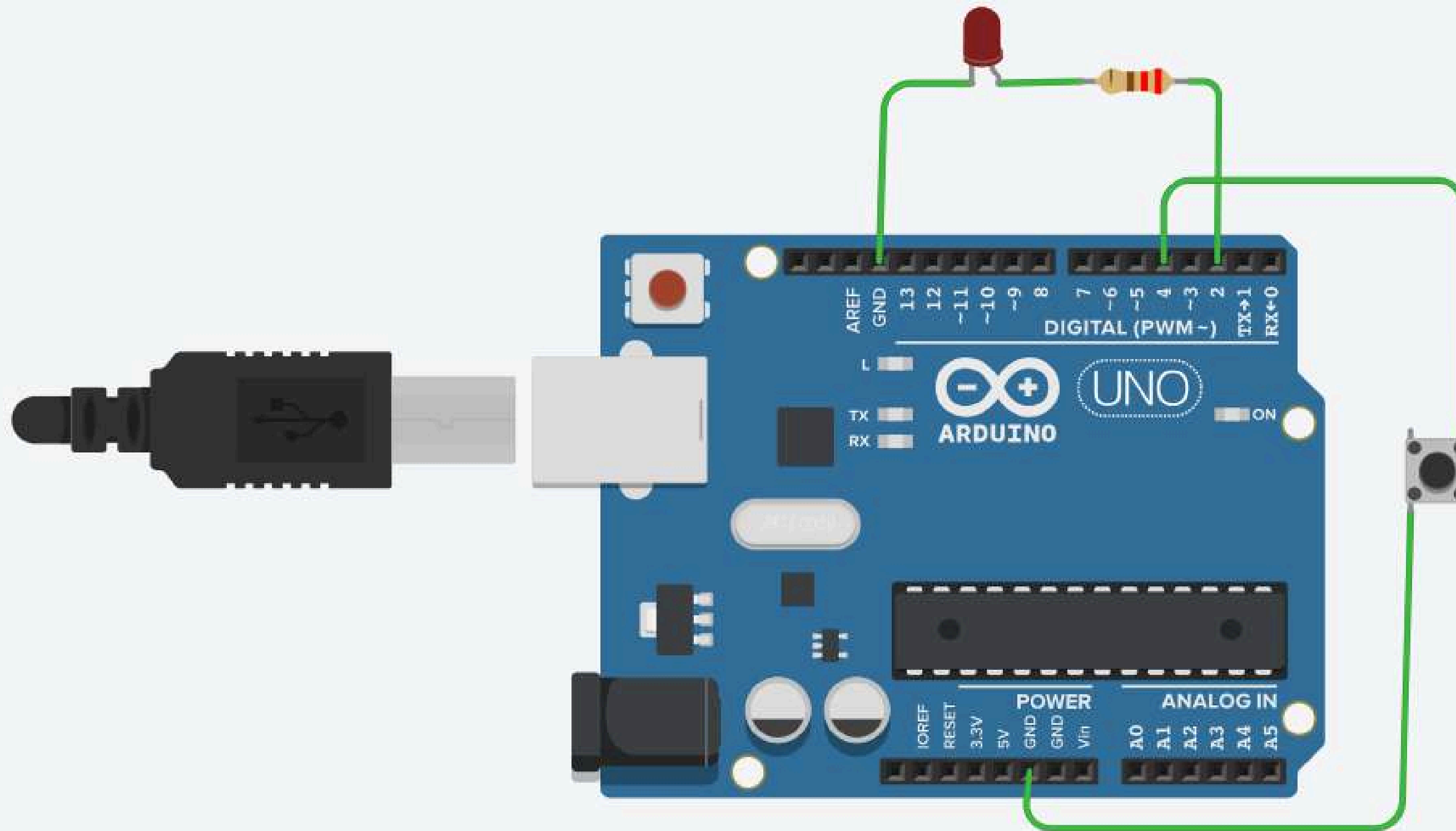
Cell_ID_A Detected
Current Zone : HOME
VALID TRANSITION

INVALID TRANSITION
ALERT GENERATED
HOME OVERSTAY DETECTED
EXPECTED TRANSITION NOT FOUND
ALERT GENERATED
HOME OVERSTAY DETECTED

Send

Clear

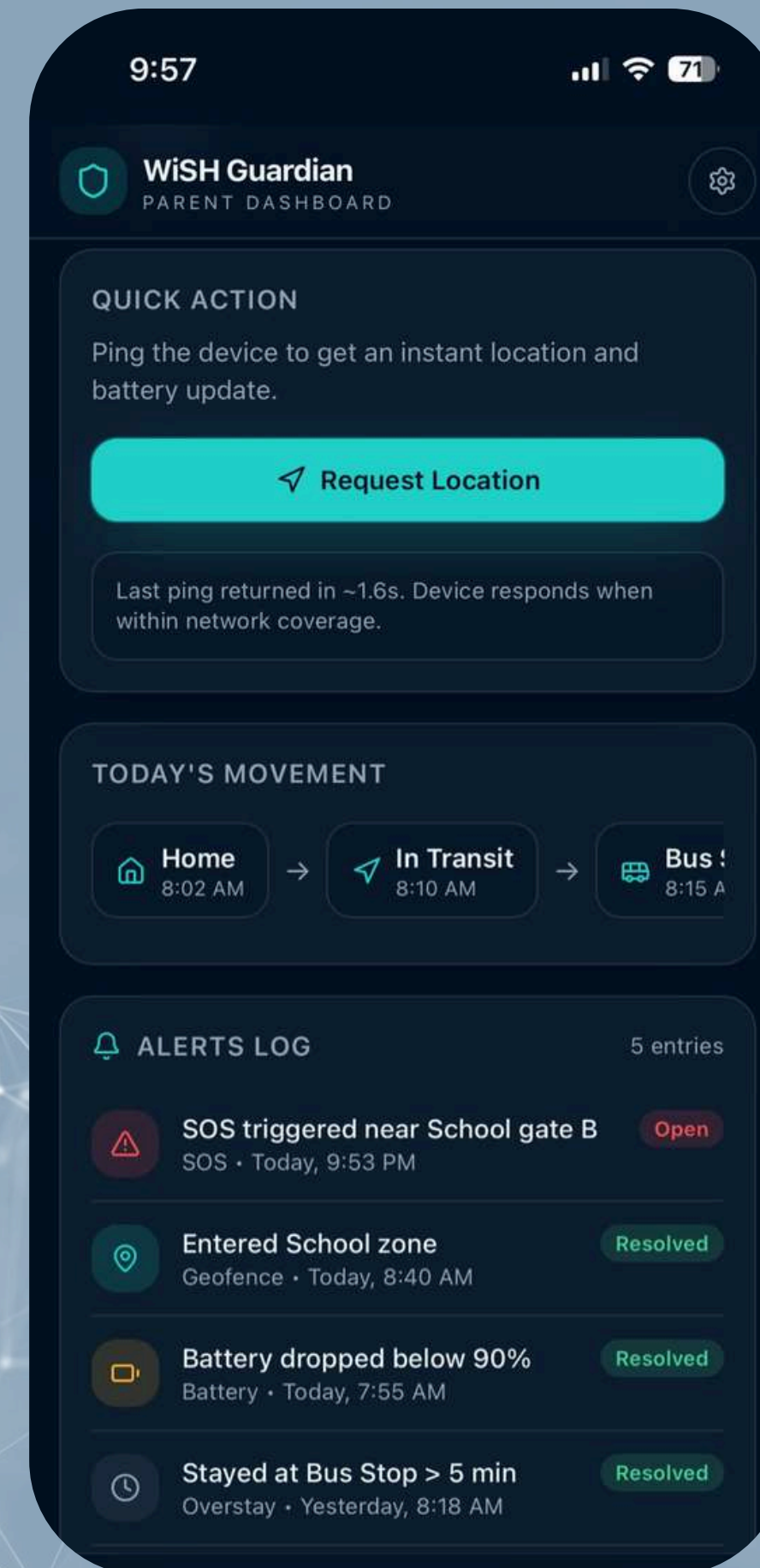
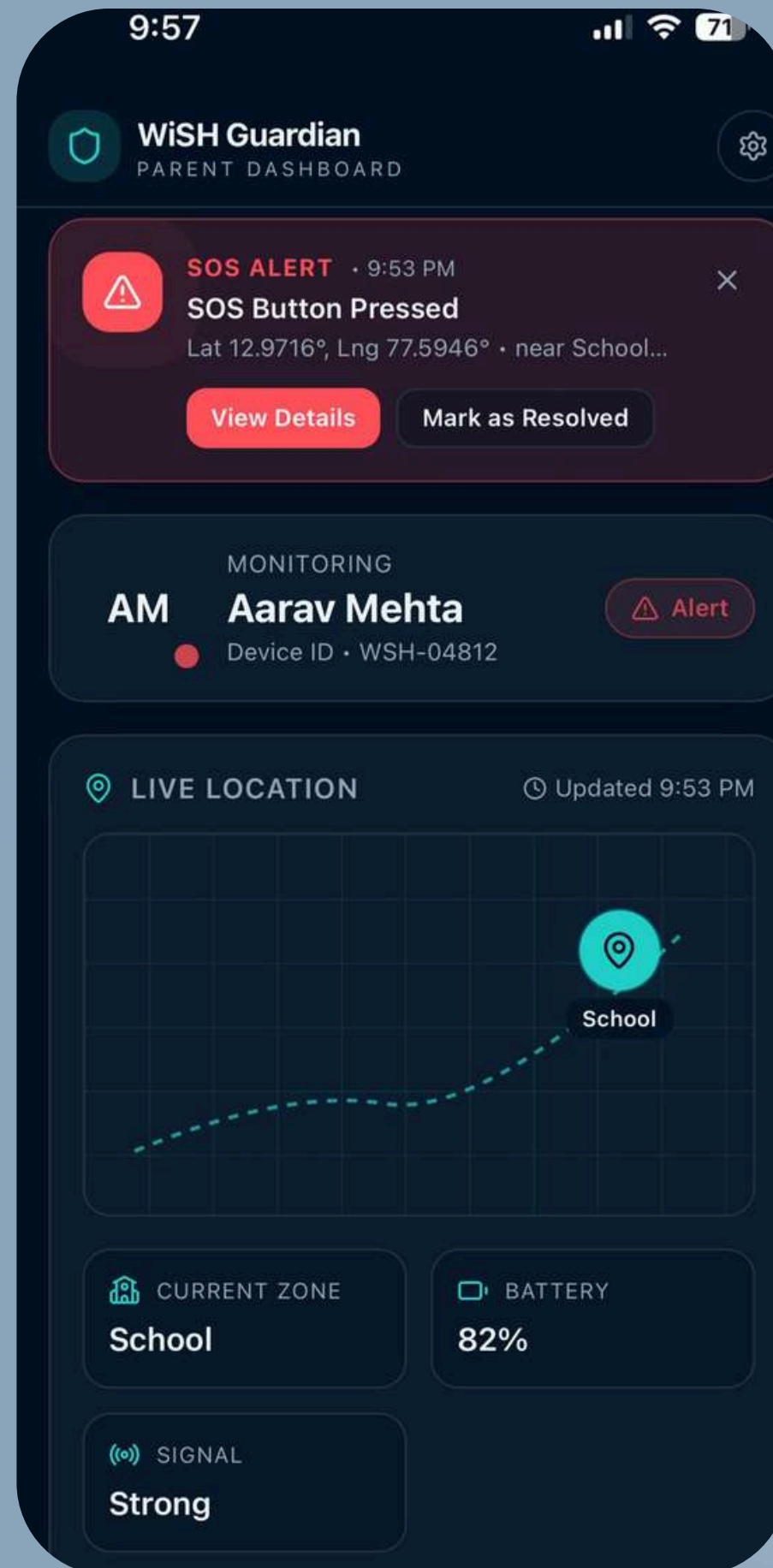
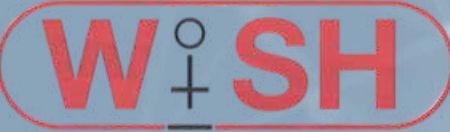
SIMULATION FOR SOS BUTTON



DEVICE WOKE UP
SOS BUTTON PRESSED!

GPS ACTIVATED
Location: 22.5726, 88.3639
ALERT SENT TO PARENT

PARENT APP INTERFACE



COST ANALYSIS

COMPONENTS	ESTIMATED COST(₹)
A7670C	500-700
ESP32	150 - 200
BQ25504	120-250
TPS62840	40-80
TPS22916C	15-30
Solar Panel	15 - 40
Supercapacitor	80 - 150
PCB	50-100
Enclosure	80-100
Antenna + Passives + Connectors	60 - 120
Waterproofing	30-50
TOTAL	1,130-1,820

TI PARTS USED

- TBQ25504
- TPS22916C
- TPS62840

FUTURE INTEGRATIONS

Using SoC

₹980-1,620

NO MCU+Size
Reduction

Using LoRA

₹1,190-1,970

NO need of signal

SoC + LoRA

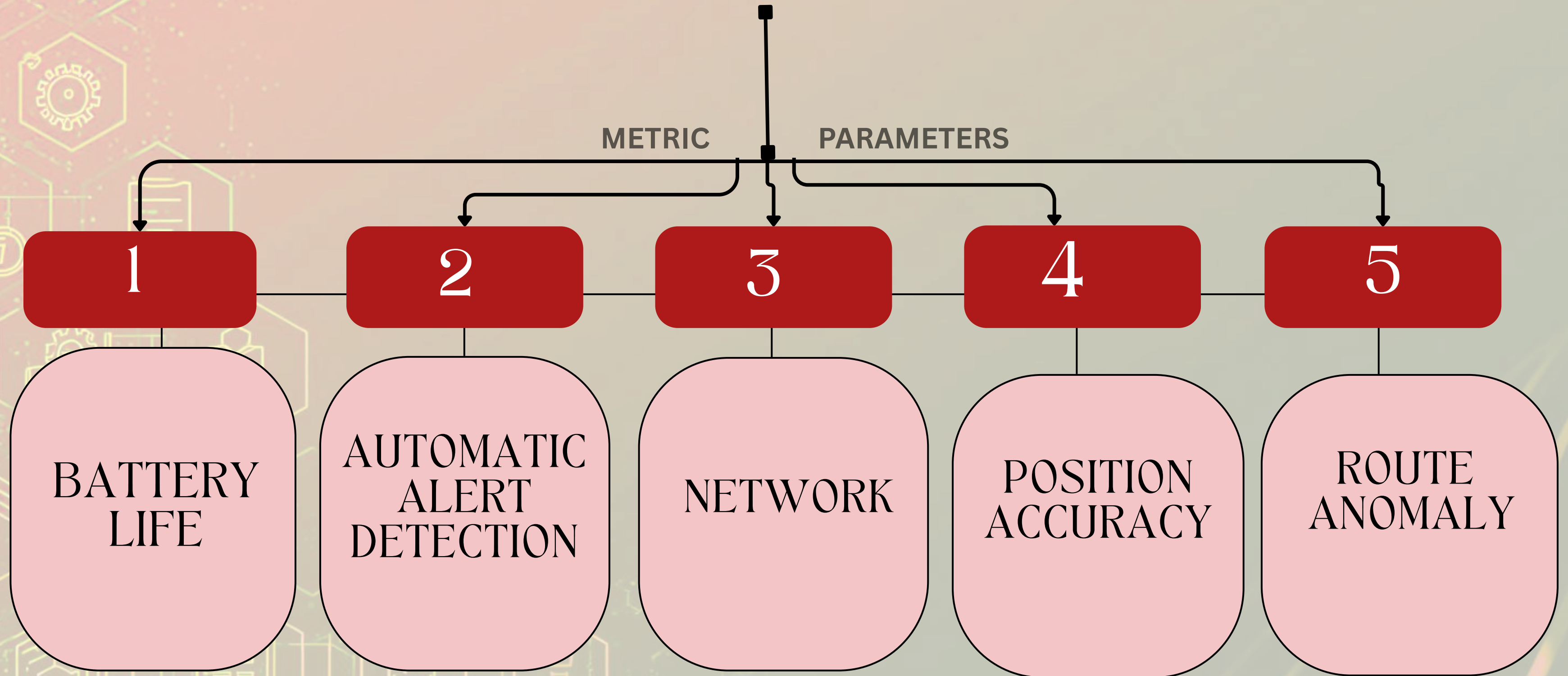
₹1,080-1,820



Step 5

Test

TESTING



SOLUTION OVERVIEW

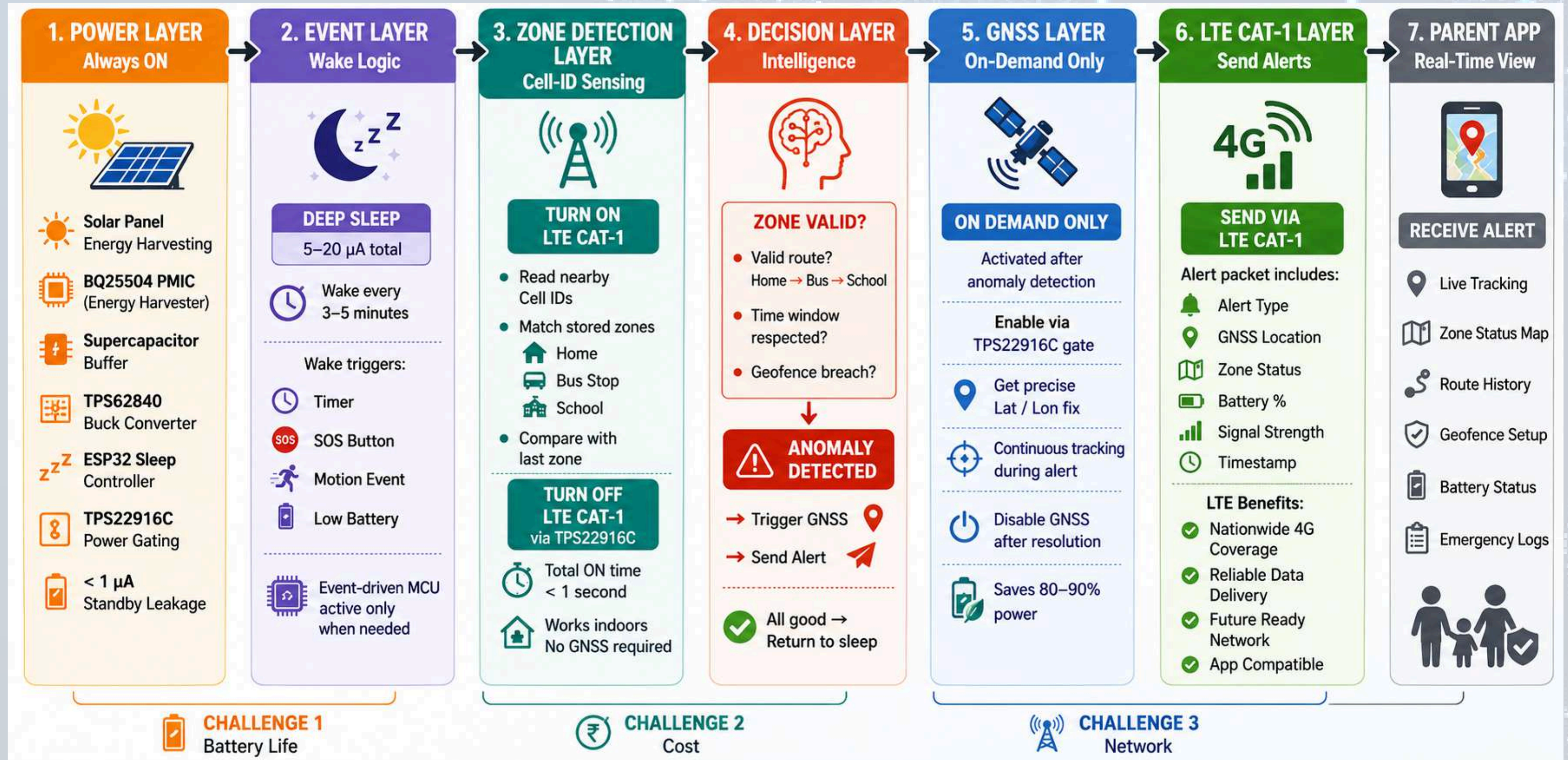


PHOTO-MOCKUP



Solar Powered
Charges during the day,
works all day.



Water Resistant
Built tough for kids.



Rugged & Safe
Designed for everyday
protection.

DIMENSIONS



KEY SPECIFICATIONS

- MCU: ESP32 (Low Power)
- Connectivity: 4G LTE Cat 1 + GNSS
- Positioning: Multi-GNSS (On Demand)
- Geofencing: Cell-ID Based + GNSS Validation
- Alerts: SOS / Anomaly via SMS, App, Dashboard
- Supercapacitor: 2.7V 10F
- Charging: Solar + USB
- Dimensions: 52mm x 32mm x 14mm
- Weight: ~35g
- Built for Kids – Light, Safe, Reliable

BUILT FOR SAFETY. DESIGNED FOR PEACE OF MIND.



Real-time
Monitoring



Smart
Geofencing



Path
Stamping



Instant
Alerts



Solar
Powered



Power
Optimized



Easy to
Wear



Privacy
Focused



School
Children



Daily
Commuters



Elderly



Spe
Nee

Tag-along – Because Their Safety is Your Peace of Mind

SUCCESS CRITERIA

**Automatic Anomaly
Detection
(Geofencing)**

**Affordable and
Scalable**

**Location Accuracy
(Multi-Layer Cell-ID +
GNSS Tracking)**

THANK YOU

WE ARE OPEN TO
QUESTIONS